ธนาคารแห่งประเทศไทย
BANK OF THAILAND

Integrating Market and Credit Risk of a Corporate Bond Portfolio

PIER Exchange

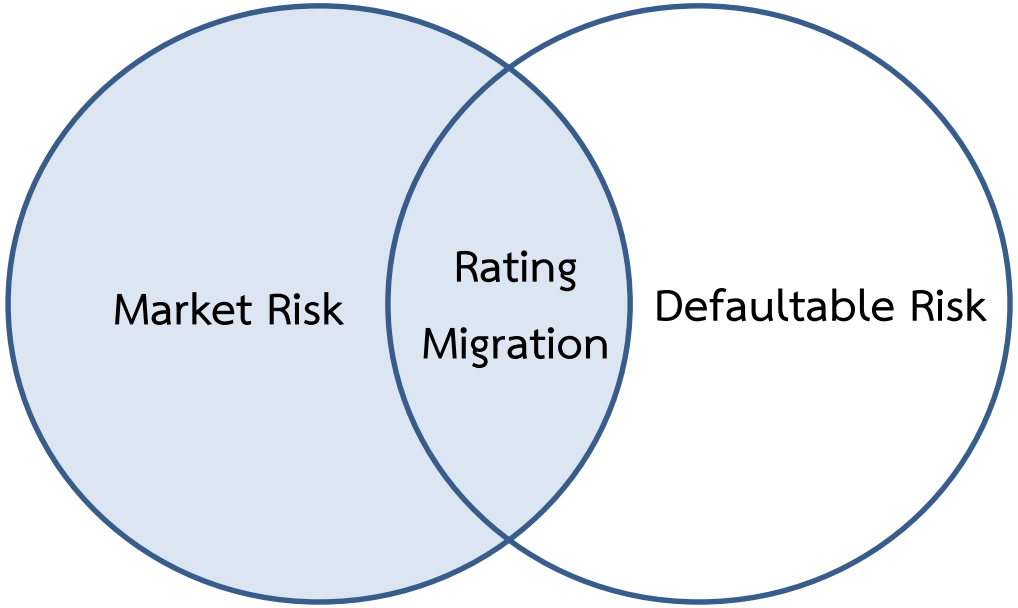
Agenda

- Introduction
- Objective
- Methodology
- Progress to date

Introduction

When there's market risk, there's credit risk:

- Corporate bonds carry both types of risk, their values are sensitive both to interest rates and to the credit worthiness of issuers.
- But, typically, their risk is quantified by market returns – mostly market risk and partly credit events (rating migration).



Introduction

“Never” does not mean “Not forever”

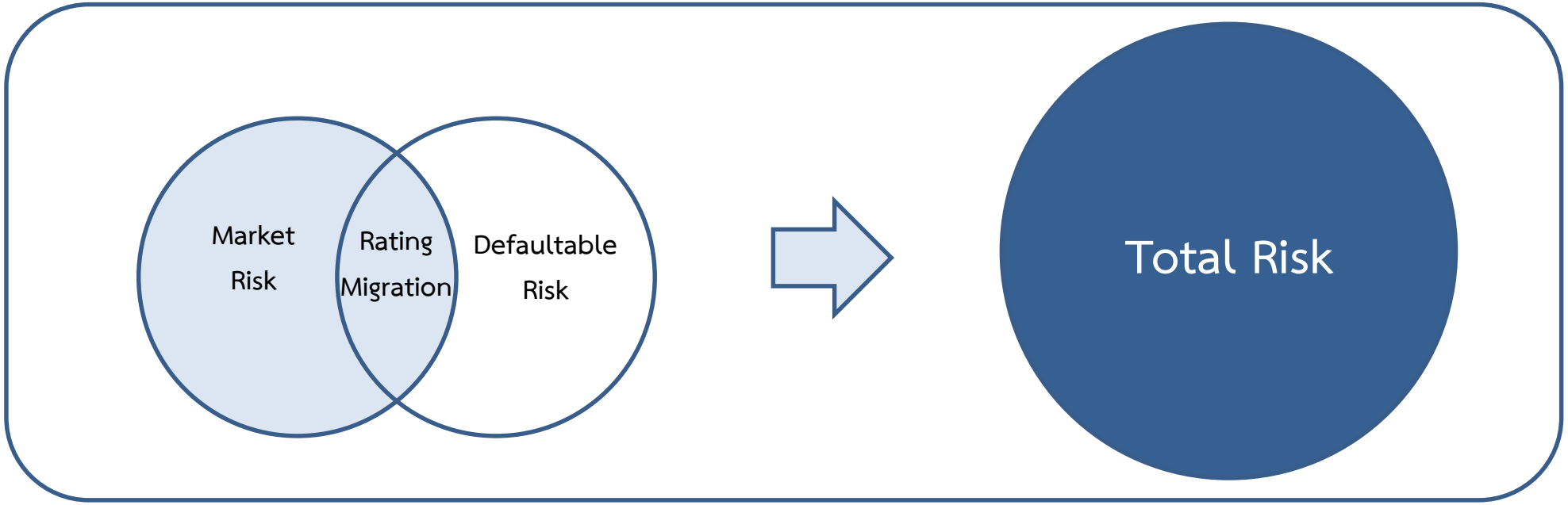
Recent Five-Year Alphanumeric Rating Migrations Rates, July 2009 Cohort

From/To:	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca	C	WR	Default
Aaa	40.21	12.37	8.25	9.28	0.00	0.00	2.06	0.00	0.00	1.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	26.80	0.00
Aa1	0.00	12.21	13.74	14.50	9.16	12.98	0.00	4.58	4.58	0.76	0.00	0.00	2.29	0.00	0.00	0.76	0.00	0.00	0.00	0.00	0.00	24.43	0.00
Aa2	0.00	1.55	5.43	28.68	10.85	14.73	5.43	4.65	5.43	1.55	0.00	0.00	0.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.93	0.00
Aa3	0.00	0.59	0.00	12.35	19.41	12.35	4.71	7.06	4.12	0.00	2.94	0.00	3.53	2.35	0.00	0.00	0.59	0.00	0.00	0.00	0.00	28.24	1.77
A1	0.00	0.00	0.00	5.91	24.41	16.93	6.69	10.24	3.94	1.97	0.79	0.39	0.79	1.18	0.00	0.00	0.39	0.39	0.00	0.00	0.00	24.80	1.18
A2	0.00	0.31	0.00	1.22	12.81	26.22	13.72	5.49	4.88	0.92	1.22	0.31	0.31	0.31	1.83	0.00	0.00	0.00	0.00	0.00	0.00	28.66	1.83
A3	0.00	0.00	0.00	0.00	1.09	8.99	30.25	12.81	7.08	1.64	1.64	0.82	0.27	0.82	0.55	0.00	0.00	0.00	0.00	0.00	0.00	32.70	1.36
Baa1	0.00	0.00	0.00	0.00	0.58	4.02	13.22	32.47	17.24	3.74	1.15	1.15	0.58	0.29	0.00	0.00	0.29	0.29	0.00	0.00	0.00	24.14	0.86
Baa2	0.25	0.00	0.25	0.00	0.00	1.52	4.04	16.16	37.37	11.87	2.02	0.76	0.51	0.00	0.76	0.51	0.00	0.25	0.00	0.00	0.00	23.49	0.25
Baa3	0.00	0.00	0.00	0.33	0.00	0.00	2.33	9.00	21.00	32.00	4.67	2.00	2.00	2.67	0.67	0.67	0.33	0.00	0.00	0.00	0.00	21.33	1.00
Ba1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.58	11.45	19.85	16.03	8.40	2.29	0.76	3.05	0.00	0.76	2.29	0.76	0.00	0.00	29.01	0.76
Ba2	0.00	0.00	0.00	0.00	0.00	0.00	0.64	0.64	3.21	11.54	11.54	21.15	8.33	3.85	3.21	1.28	0.64	0.00	0.00	0.00	0.00	32.05	1.92
Ba3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.55	2.59	4.66	7.77	7.77	11.92	7.77	3.11	2.07	2.59	0.52	0.00	0.00	0.52	45.08	2.07
B1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.45	2.26	0.45	4.07	8.60	15.39	7.24	0.91	2.26	0.45	0.91	0.45	0.00	53.85	2.72
B2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.89	0.76	2.26	3.77	7.55	9.81	9.06	4.15	2.64	0.38	1.13	0.76	50.19	5.66
B3	0.00	0.00	0.00	0.00	0.00	0.00	0.27	0.00	0.00	0.55	0.55	0.82	1.09	2.73	6.01	10.93	8.74	1.37	0.82	0.27	0.00	59.56	6.28
Caa1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.86	0.43	0.00	1.29	2.16	5.60	9.91	11.64	3.02	1.29	0.00	0.00	47.41	16.38
Caa2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.75	0.00	0.00	1.49	2.99	8.21	3.73	7.46	0.00	0.00	0.75	39.55	35.08
Caa3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.47	0.00	0.00	2.94	2.94	5.88	0.00	1.47	0.00	35.29	50.00
Ca	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.64	0.00	3.28	0.00	0.00	0.00	32.79	62.30
C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.67	0.00	0.00	0.00	0.00	0.00	0.00	13.33	80.00

Source: Moody’s

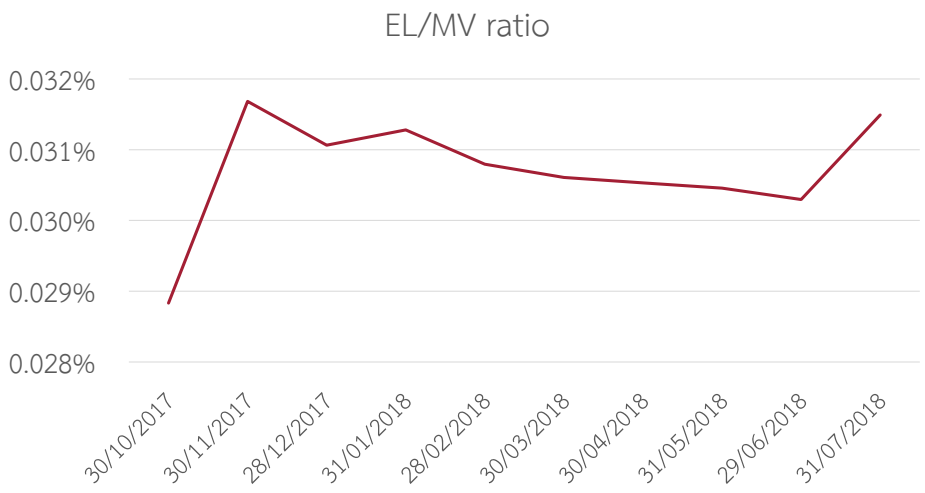
Objectives

- Quantify defaultable credit risk at firm-level of US and EU corporate bonds.
- Aggregate the correlated types market (observed) and credit (quantified) risk into total risk
- Applications

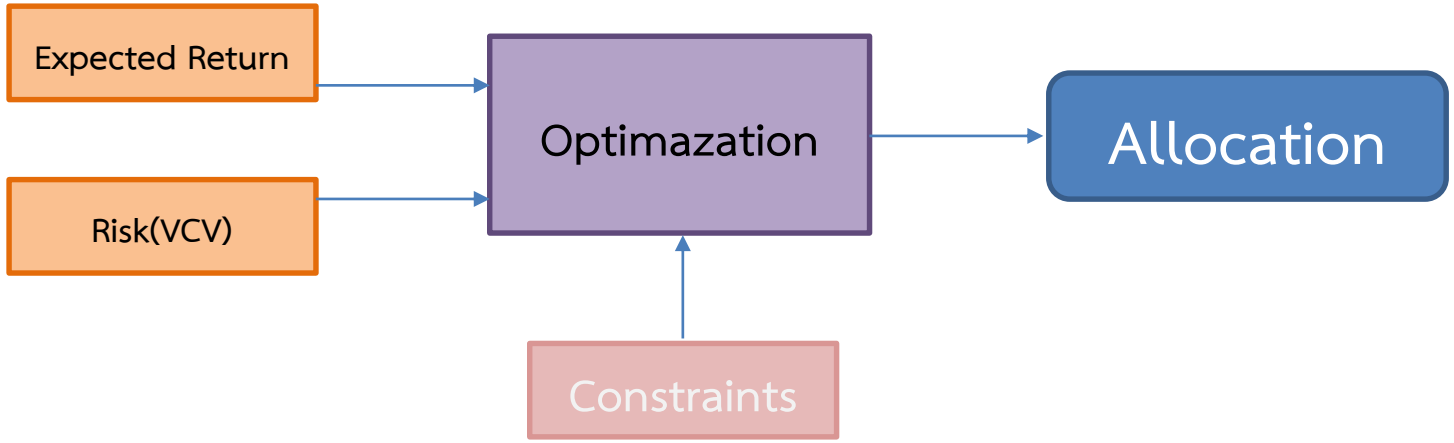


Application

Risk monitoring for corporate bond portfolios



Portfolio construction



Data

- US and EU corporates according to BOT's monthly customized indices from Barclays Point from 2000 to 2017 (returns and firm characteristics)
- Monthly firms' balance sheets from Bloomberg from 2000 to 2017
- Monthly firm stock returns from Bloomberg from 2008 to 2017
- US and EU interest rates from RiskMetrics from 2000 to 2017

	AAPL	AAPL	ABBNVX	ABBNVX	ABIBB	ABIBB	ABT	ABT	ADM	ADM
	Market Valu	Maturity	Market Valu	Maturity	Market Valu	Maturity	Market Valu	Maturity	Market Valu	Maturity
30/06/2015	4.16	6.06	0.26	5.4	3.21	4.48	0.63	6.01	0.24	4.23
31/07/2015	4.18	5.98	0.27	5.33	3.19	4.4	0.64	5.93	0.2	3.85
31/08/2015	4.17	5.9	0.26	5.24	3.17	4.32	0.63	5.85	0.2	3.76
30/09/2015	4.22	5.83	0.27	5.17	3.17	4.23	0.63	5.78	0.18	3.86
30/10/2015	4.19	5.75	0.26	5.09	3.06	4.22	0.63	5.71	0.17	3.78

Methodology

Expected defaultable credit loss = PD x LGD x EAD

To calculate the expected defaultable credit loss at firm-level:

- We apply Merton model to estimate the PDs and LGDs.
- For each firm, EADs are depicted as a distribution of bond prices.
- Prices of corporate bonds are driven by an interest rate and a credit spread. Hence, some models are required.

Methodology

To calculate the credit loss at firm-level: PD

- Merton model assumes that a firm is funded by issuing a zero-coupon bond and shares:

$$A_t = D_t + E_t,\ 0 \le t \le T.$$

- The firm defaults if the asset value falls below the face value of debt at the maturity, that is,

$$A_T \le D$$

- Merton's model relates the market value of assets, equity, and liabilities in an option pricing framework
- The asset of firms follows geometric Brownian motion

$$A_t = A_0 e^{\left(\mu_A - \frac{1}{2}\sigma_A^2\right)t + \sigma_A^2 W_t}$$

- Contingent claim:

$$E_t = \max(A_t - D)$$
$$D_t = De^{-r(T-t)} - \max(0, D - A_t)$$

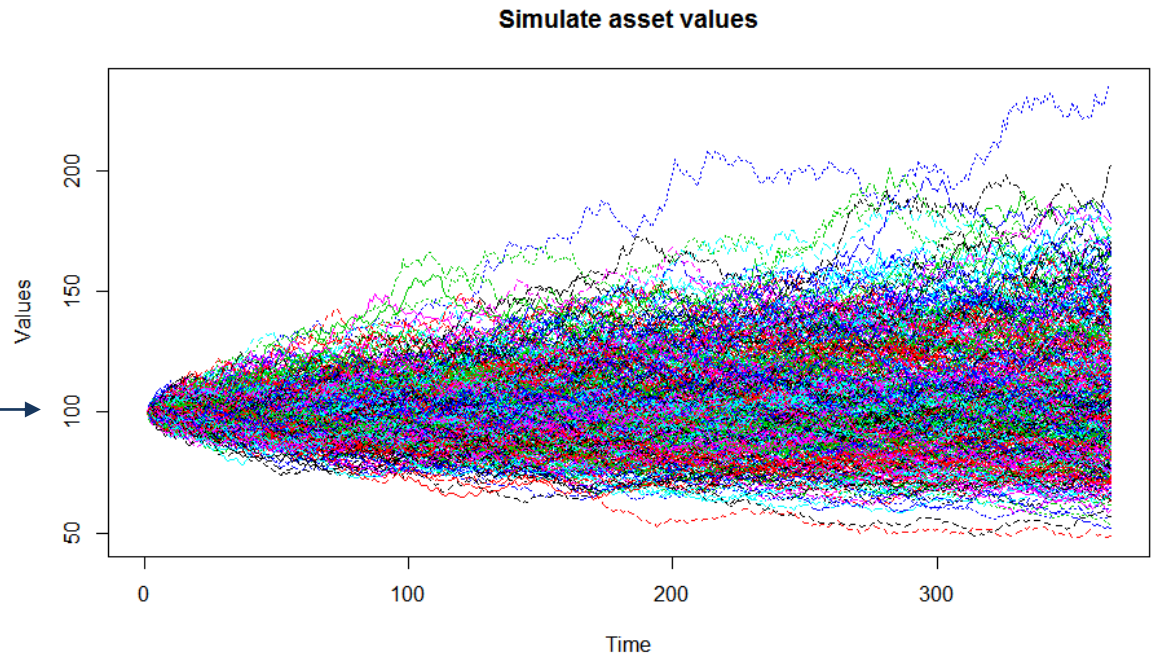
Methodology

To calculate the defaultable credit loss at firm-level: PD

- A_t follows the following diffusion process:

$$A_t = A_0 e^{\left(\mu_A - \frac{1}{2}\sigma_A^2\right)t + \sigma_A^2 W_t}.$$

$A_0 = 100$



Methodology

To calculate the credit loss at firm-level: PD

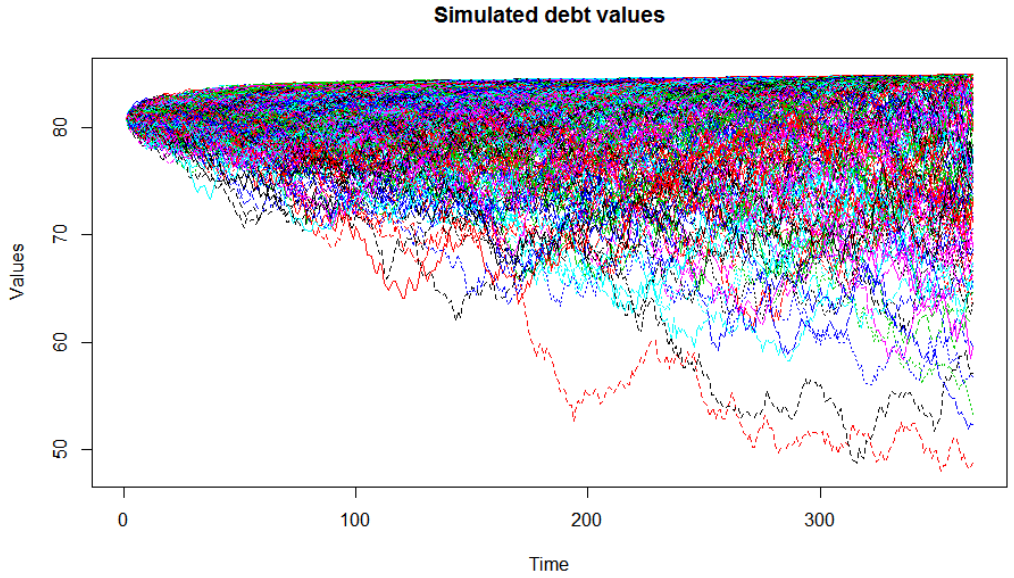
- D_t follows the following duffusion process:

$$D_t = \boxed{De^{-r(T-t)}} - \boxed{\max(0, D - A_t)}$$

a put option

deterministic, non-defaultable

stochastic, defaultable

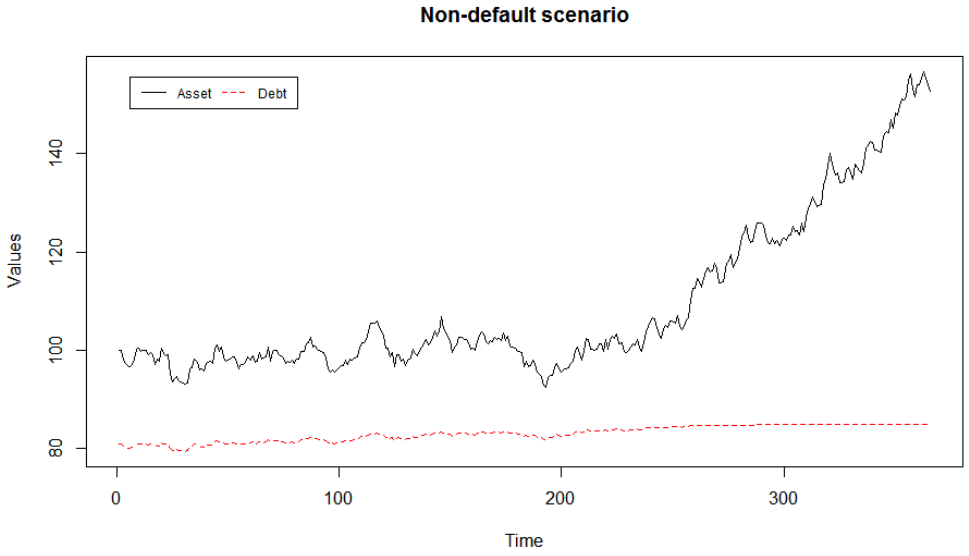


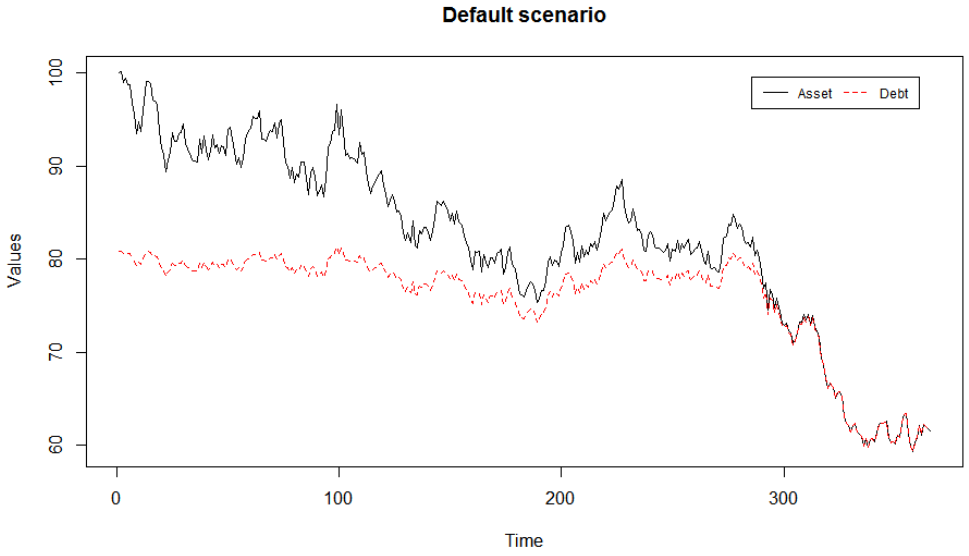
face value = 85

Methodology

To calculate the credit loss at firm-level: PD

- If a firm fares well, it can pay debt at the maturity. But in some adverse scenarios, it will end up in default ($A_T \leq D$).
- The firm asset value A_t is more likely to dip if μ_A is low and σ_A is high.

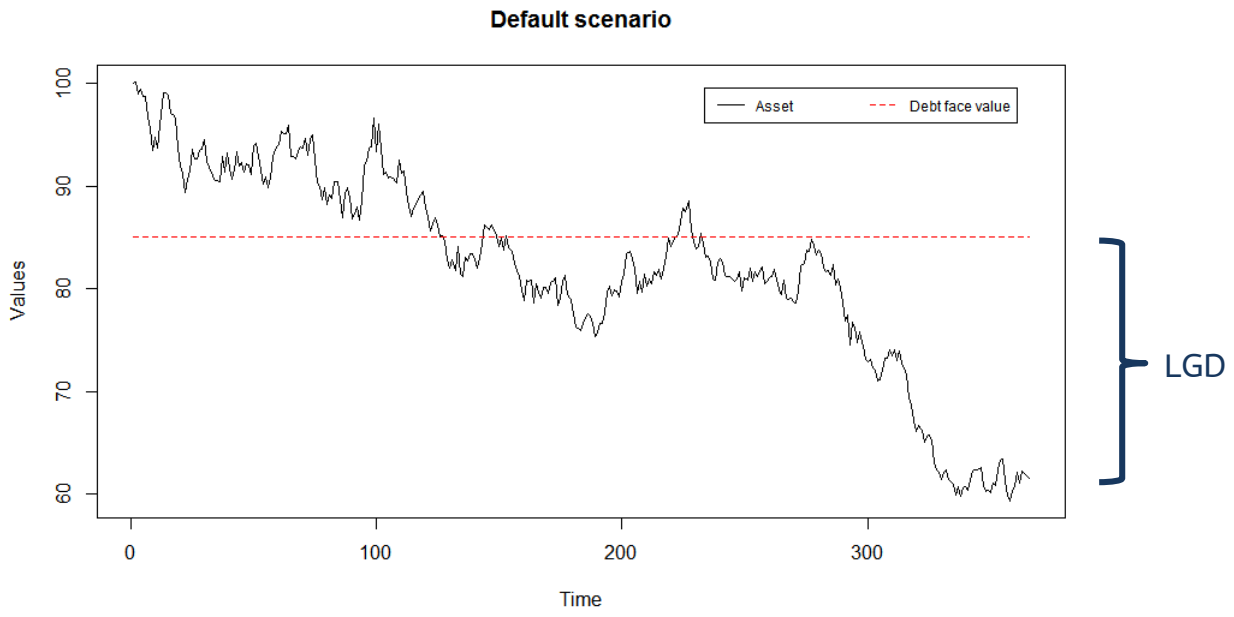




Methodology

To calculate the credit loss at firm-level: LGD

- LGD is calculated from the difference of the terminal asset value and the debt face value at the maturity, i.e., $D - A_T$.



Methodology

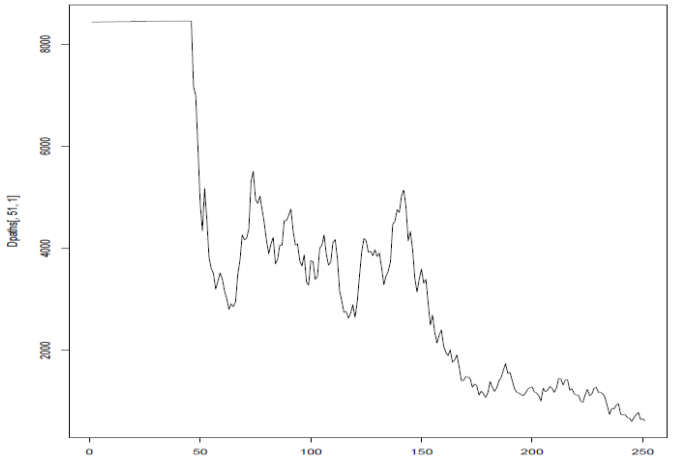
To calculate the credit loss at firm-level: EAD

- EADs are simply bond values at the maturity under various scenarios.
- They can be also illustrated as a distribution of bond values.
- Factors determining the bond values of each firm are:
  - The country's interest rates (sovereign yield curve)
  - The credit spread of each firm (an average spread over the horizon and assumed to be the same across tenors)

Methodology

To calculate the credit loss at firm-level:

- We employ Vasicek model to estimate the dynamics of the short rate and the term structure → an evolution of yield curves.
- The credit spread of each firm is implied from the simulated bond yields obtained from Merton model, deducted by the risk-free rate.



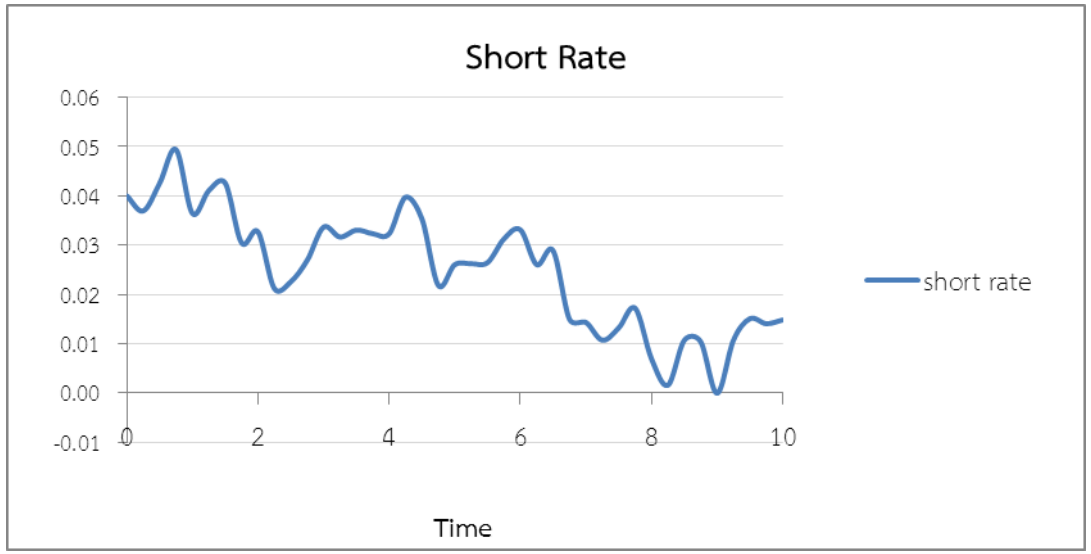
- The credit spread is then added up to the yield curves derived from Vasicek model to calculate bond prices under various scenarios → firm-level EADs.

Methodology

To calculate the credit loss at firm-level: Sovereign yield curve

- The interest rate dynamics and the term structure follow Vasicek model:
  - The short rate model (overnight perhaps)

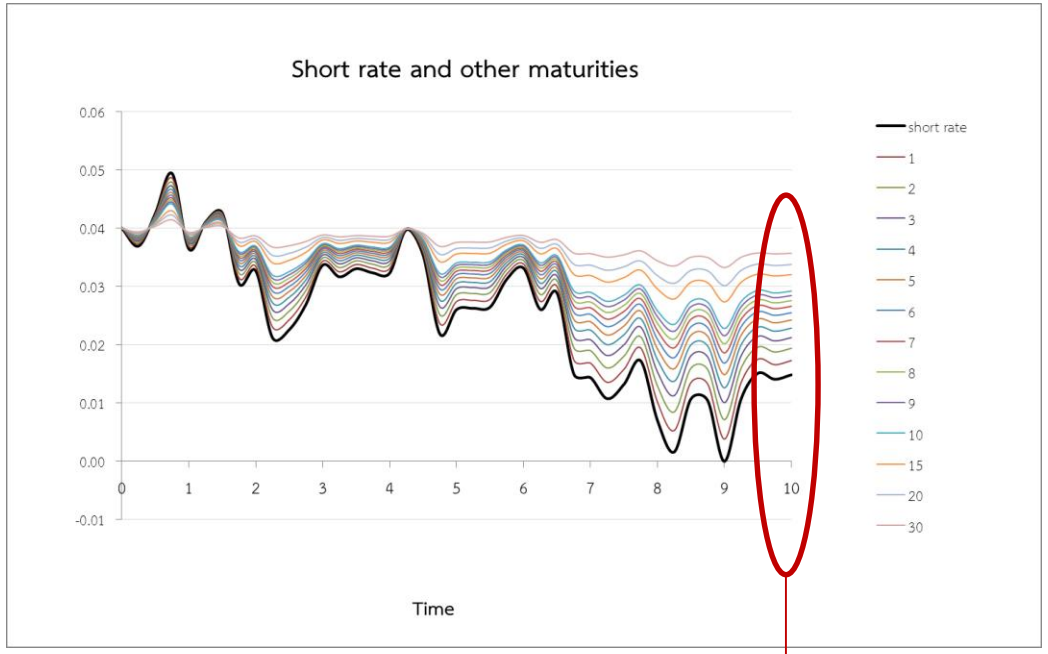
$$r_t = r_{t-1} + \alpha(\gamma - r_{t-1})\Delta t + \sigma\sqrt{\Delta t}W_t$$

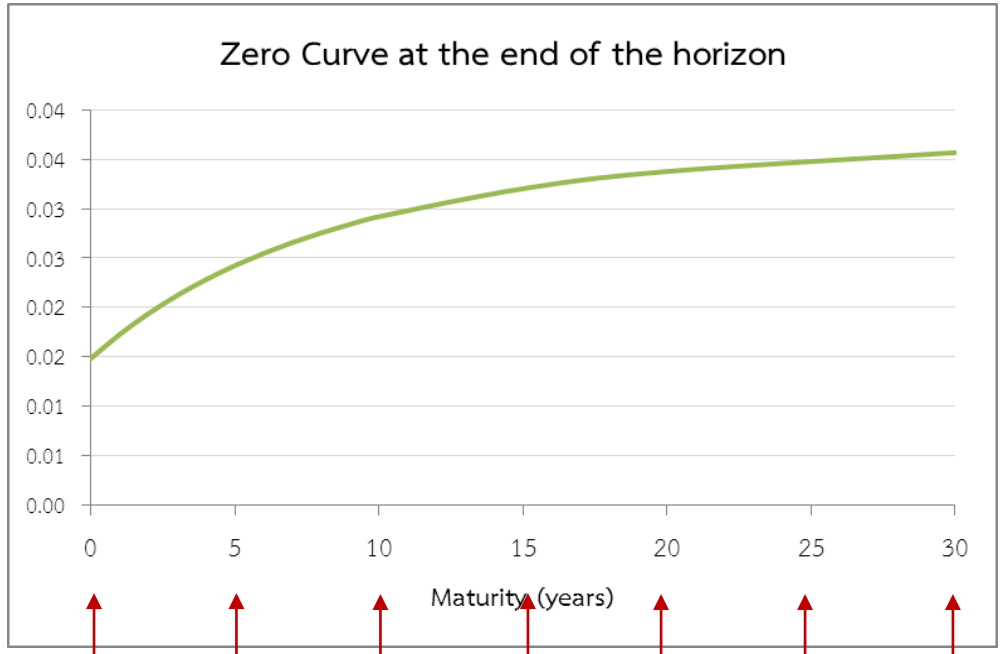


Methodology

To calculate the credit loss at firm-level: Sovereign yield curve

- Find the yield curve at the end of the horizon.





Methodology

To calculate the credit loss at firm-level: EAD

- Discount curve = yield curve + credit spread

$$D_t = De^{-(r_t^{\tau} + credit_spread)\tau}$$

where

τ is the remaining maturity of a bond at the end of the horizon,

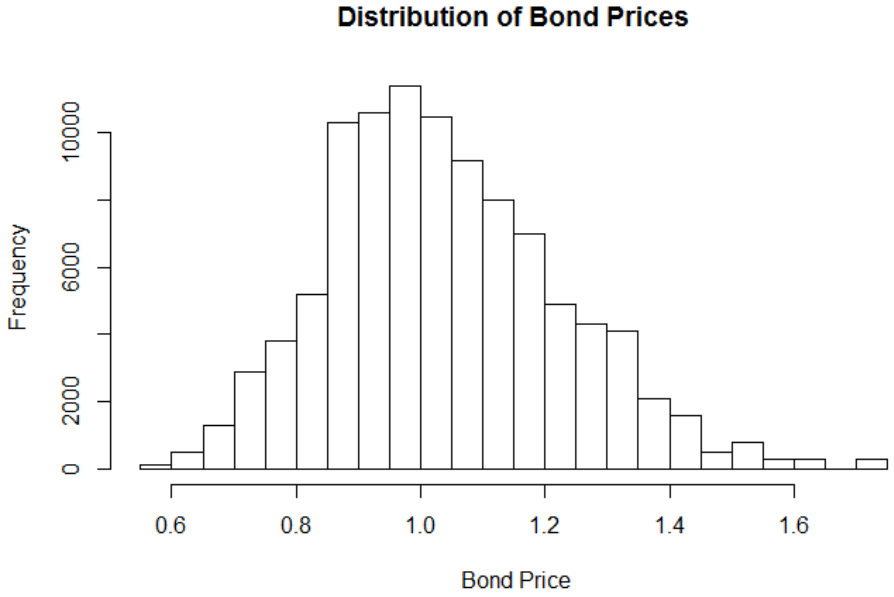
$credit_spread$ is the firm-specific credit spread,

so the discount curve is different from firm to firm.

Methodology

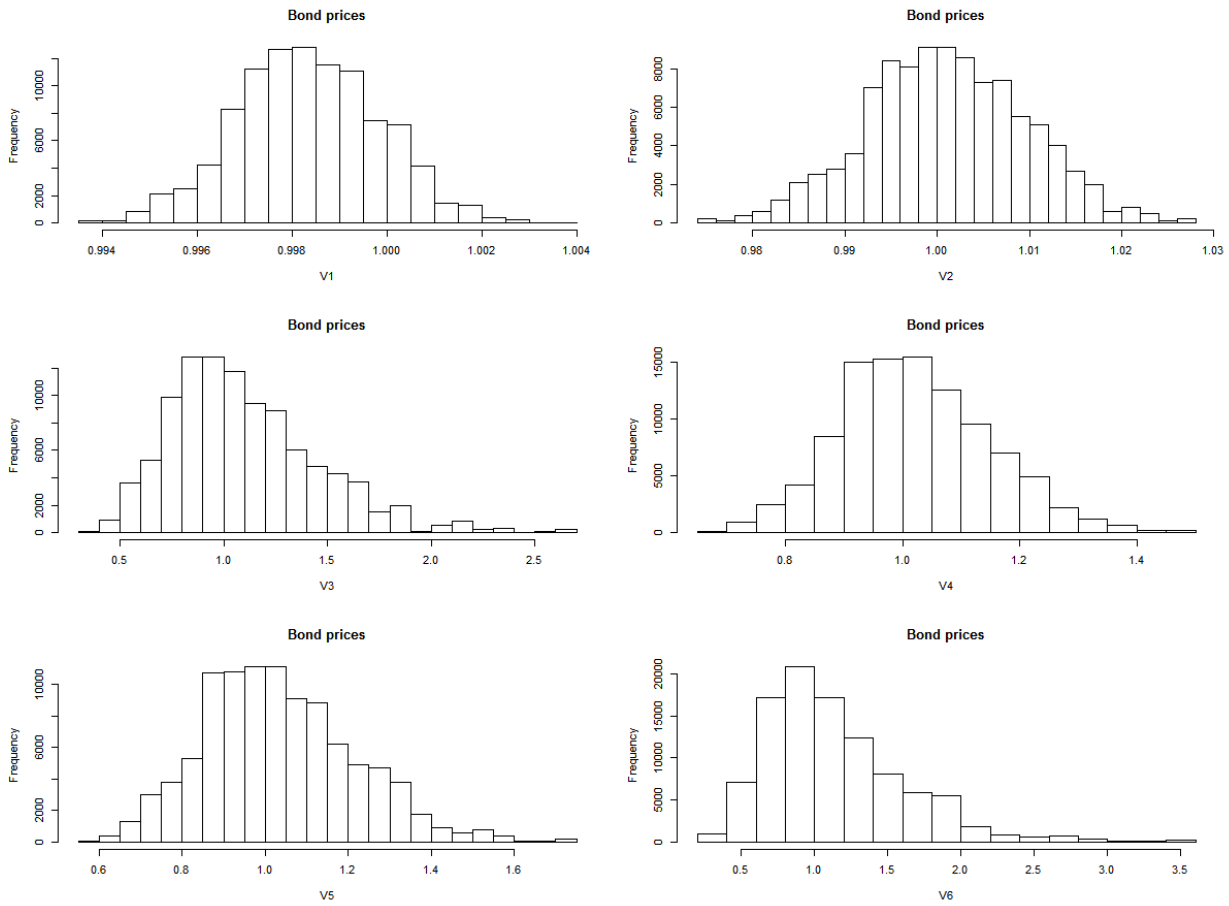
To calculate the credit loss at firm-level: EAD

- A bond is priced according to the rates simulated (as a zero-coupon bond).



Methodology

To calculate the credit loss at firm-level: EAD of each firm



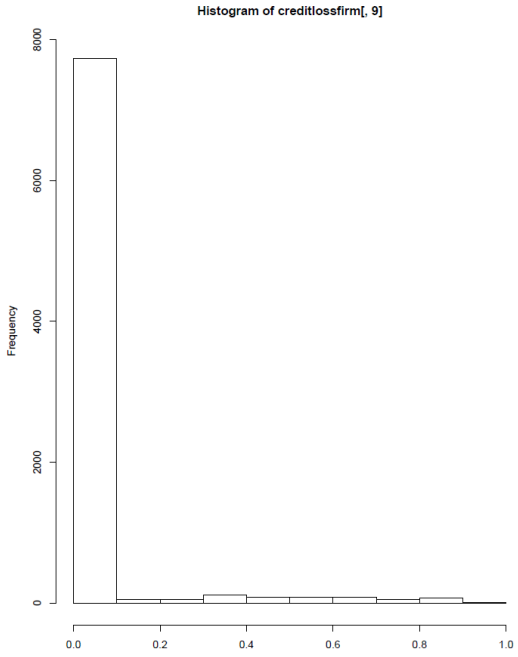
Methodology

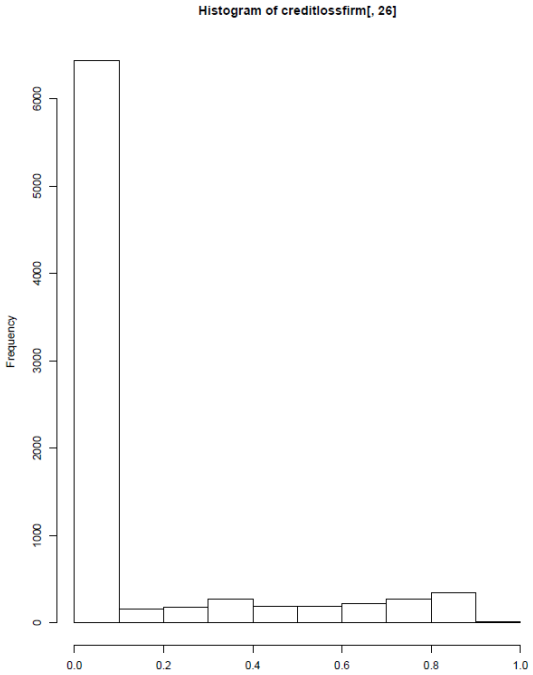
To calculate the credit loss at firm-level: combining PD, LGD and EAD

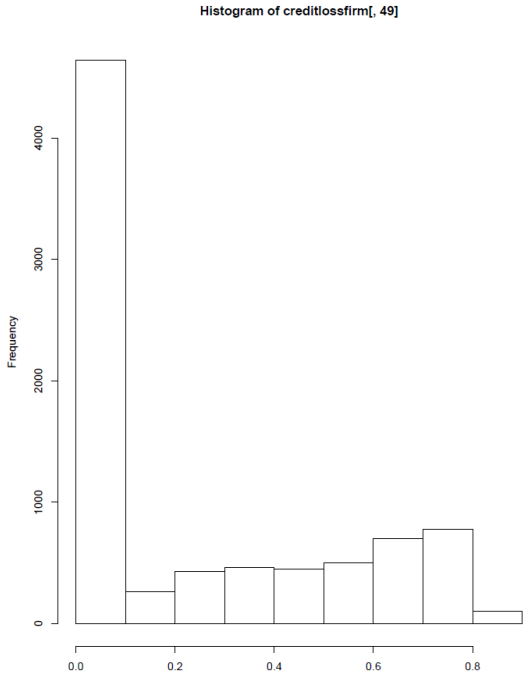
- For each firm, we have:
  - PD, as a percentage (some firms may have PD = 0)
  - LGD, as a value
  - EAD, as a distribution
- We can then calculate a loss distribution firm-by-firm.

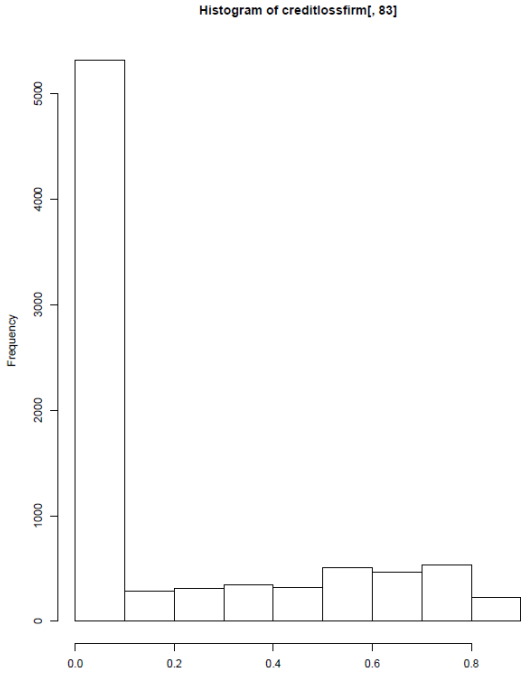
Methodology

To calculate the credit loss at firm-level: combining PD, LGD and EAD

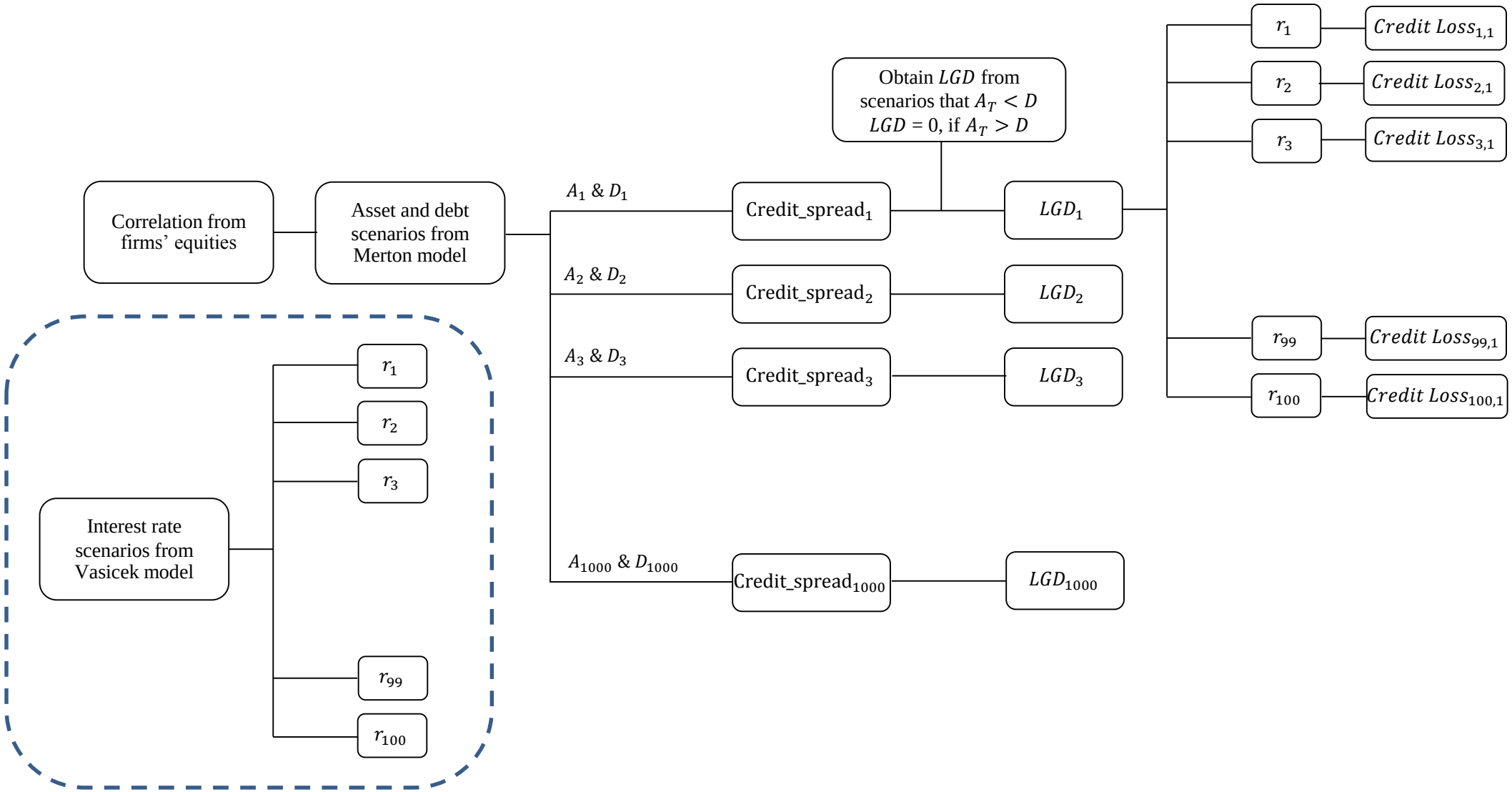




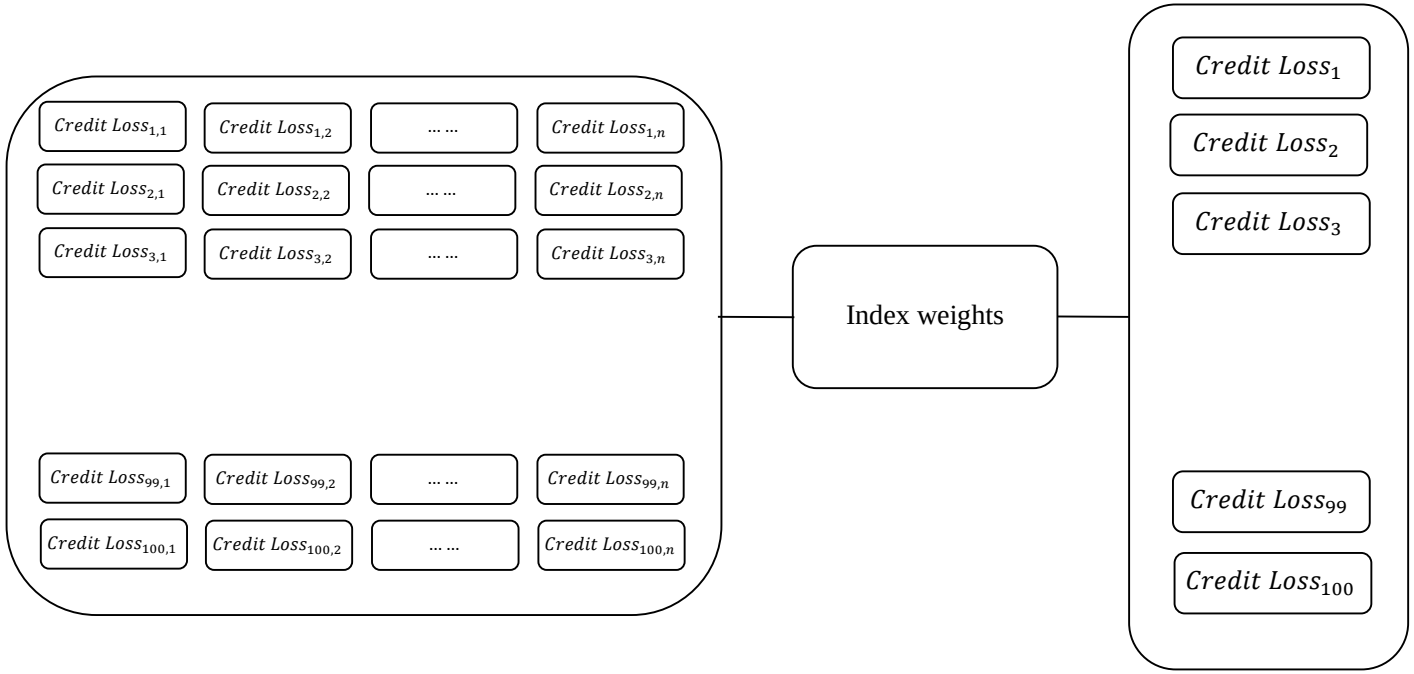


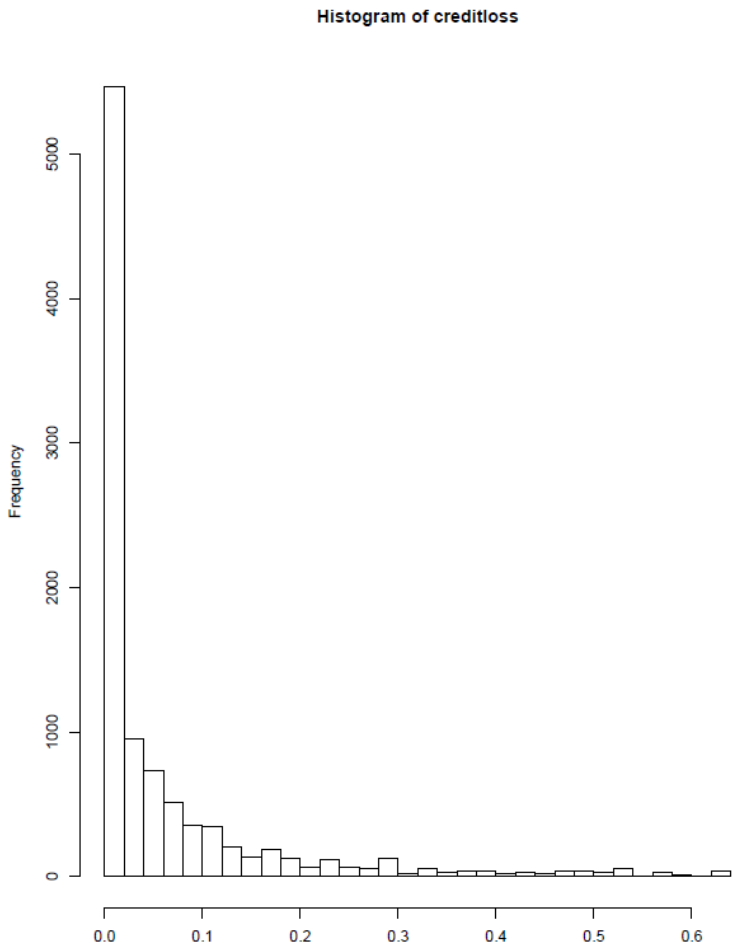


Schematic Diagram



Schematic Diagram





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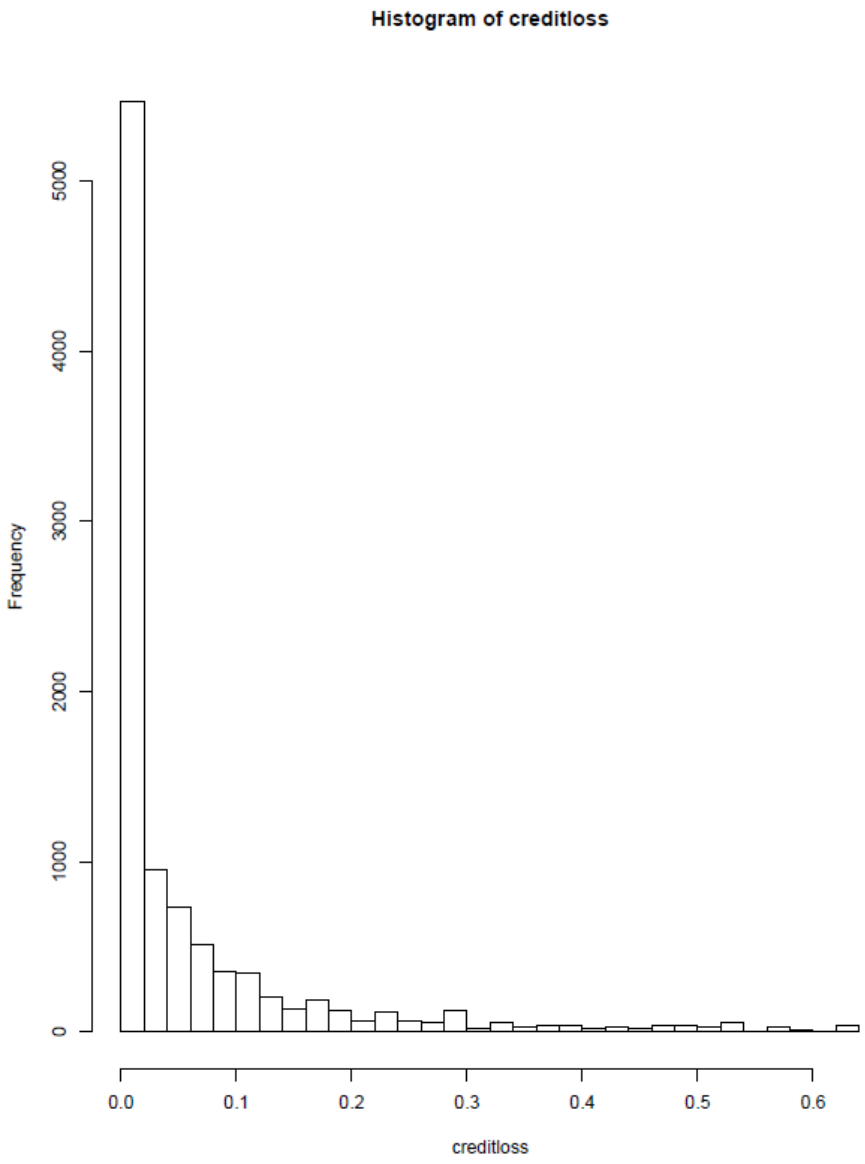
BANK OF THAILAND

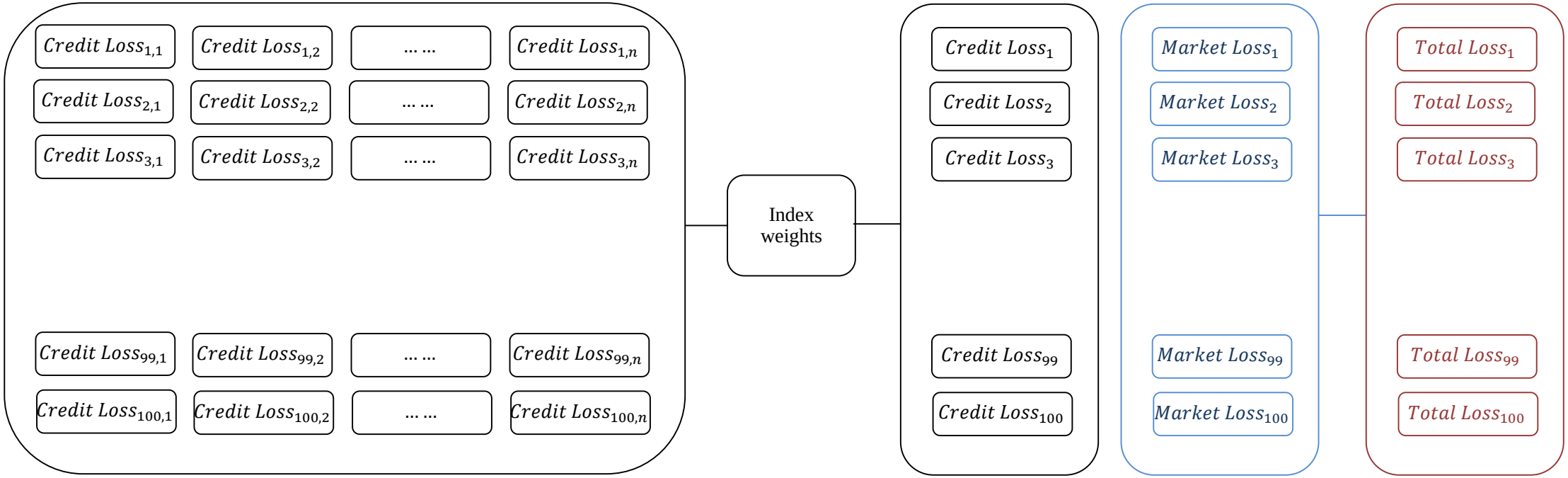
Credit Loss Distribution



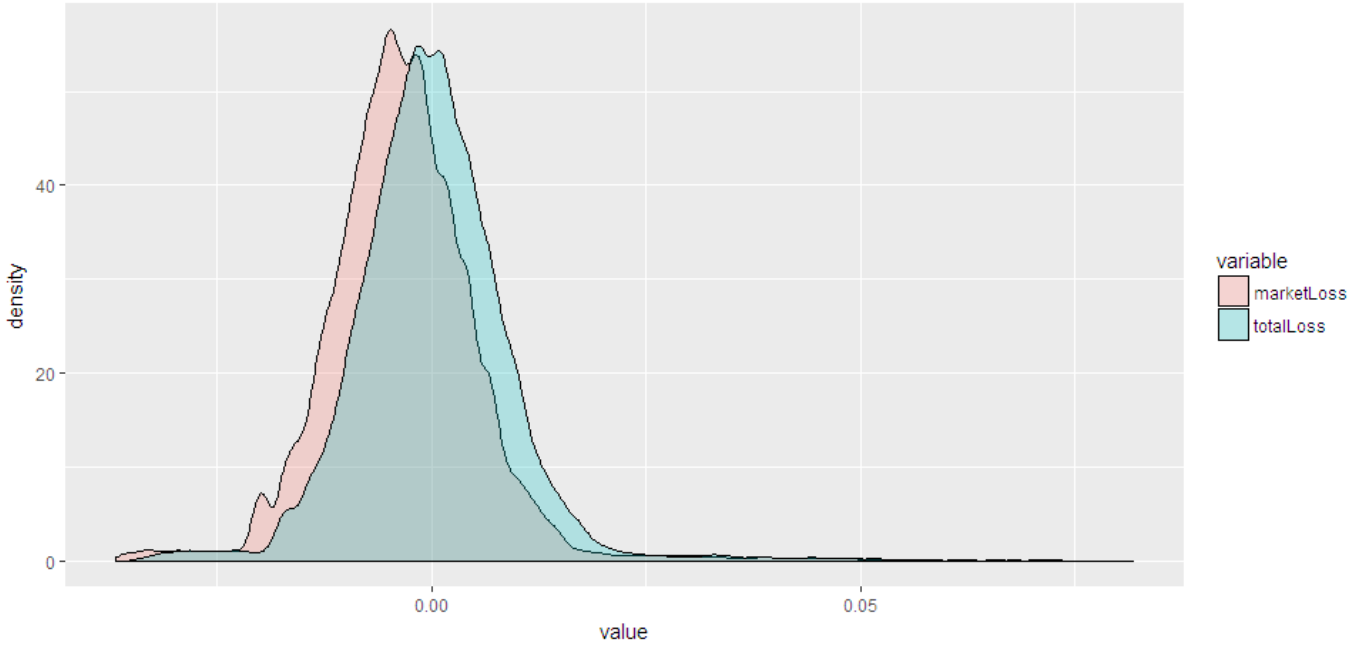
What's in progress?

- Fit market monthly returns by normal diffusion process
- Simulate market returns from correlated Wieners
- Combine the market returns with credit loss

Schematic Diagram



Goal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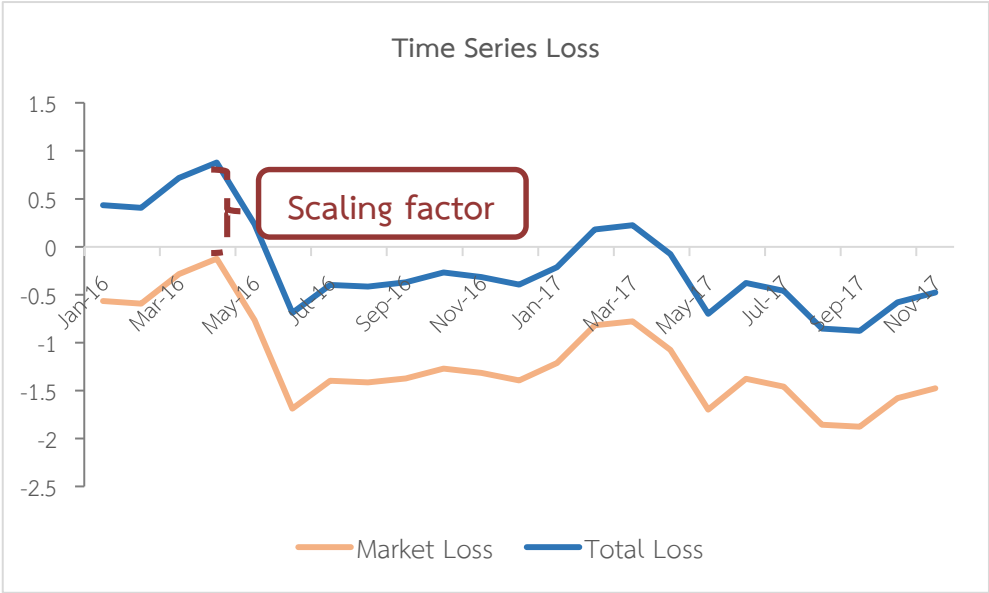
20
0
0.00
0.05
value
variable
marketLoss
totalLoss


Time Series Loss
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
Scaling factor
Jan-16
Mar-16
May-16
Sep-16
Nov-16
Jan-17
Mar-17
Sep-17
Market Loss
Total Loss